Training Day 5

1. Topics Covered and Brief Explanation

1. Conditional Statements

Conditional statements are used to execute specific blocks of code based on given conditions. Studied if, elif and else statements to control program execution.

Example:

```
marks = 75
if marks >= 90:
    print("Grade A")
elif marks >= 60:
    print("Grade B")
else:
    print("Grade C")
```

2. Looping Statements

Loops are used to execute a block of code repeatedly. Learned about two main types of loops:

- for loop: Used to iterate over a sequence.
- while loop: Executes repeatedly as long as a condition remains true.

Example:

```
for i in range(1, 6):
    print(i)


count = 1
while count <= 5:
    print(count)
    count += 1
```

3. Functions in Python

Functions are reusable blocks of code designed to perform specific tasks. Learned how to define functions using the def keyword, pass arguments and return values.

Example:

```
def add(a, b):
```

```
    return a + b


result = add(10, 20)
print("Sum:", result)
```

4. Python Data Structures

Studied different data structures used to store and organize multiple values.

Data Structure	Description	Example
List	Ordered and changeable collection.	[10, 20, 30]
Tuple	Ordered and unchangeable collection.	(10, 20, 30)
Dictionary	Stores data in key-value pairs.	{"name": "Neha"}
Set	Stores unique elements.	{10, 20, 30}

5. Basic Operations on Data Structures

Learned operations such as adding, removing, accessing and updating elements in lists and dictionaries.

2. Tools and Technologies Used

Tool	Purpose
Python 3	Writing and executing Python programs.
Google Colab	Practising Python code in an online environment.
Google Chrome	Accessing Python documentation and learning resources.

3. Practical Learning / Code Examples

Example 1: Working with Lists

```
numbers = [10, 20, 30, 40]
numbers.append(50)
numbers.remove(20)
print("List:", numbers)
print("First Element:", numbers[0])
```

Example 2: Working with Dictionaries

```
student = {
    "name": "Navneet",
    "age": 21,
    "marks": 95
}

print(student["name"])
student["marks"] = 98
print(student)
```

4. Learning Outcomes

- Understood the use of conditional statements in Python.
- Learned to use for and while loops.
- Gained knowledge of creating and calling functions.
- Understood lists, tuples, dictionaries and sets.
- Practised writing structured and reusable Python programs.